# Daniil Zinovyev

Senior Software Development Engineer

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Distributed systems engineer: high-throughput serverless services, the multi-region infrastructure they ship on, and AI developer tooling.

## EXPERIENCE

### Senior Software Development Engineer, Amazon — March 2021 – Present

*Internal learning platform training 2M+ frontline workers a year across ~2,000 sites. Roughly 24 services and 40 deployment pipelines live in 4 AWS regions. Promoted twice: SDE I to SDE II to Senior.*

- Co-led the replacement of a 15-year-old tier-1 identity service serving 280 client systems at a 5,000 request/second baseline, a deprecation deferred since 2021. Reached full feature parity in four weeks and consolidated 22 legacy RPC operations into 7 on Smithy/CBOR, over Lambda, DynamoDB and OpenSearch Serverless.
- Built that service's resilience layer: circuit breakers on every dependency, per-client rate limiting on ElastiCache/Valkey with dynamic configuration, and chaos and fault-injection testing. Sustains 5,000 req/s with over 2x headroom while down an availability zone, validated by capacity modelling with Little's Law. Wrote the 1,200-line handbook: SLOs, a runbook per alarm, and disaster recovery at a 120-minute RTO and 5-second RPO.
- Built the platform's multi-region CI/CD from nothing: region-ordered production waves, bake times, promotion gated on the receiving region's own service health, automatic rollback on production alarms, and deploys blocked automatically during company-wide peak traffic events.
- Shipped 7 internal AI developer tools adopted beyond my team: a Model Context Protocol server exposing platform APIs to coding agents (~20 tools, 988 tests), and a multi-agent review pipeline whose reviewers are read-only by construction, so the agent that wrote the code cannot approve it. An AWS Bedrock (Claude) migration tool is projected to cut per-service region migration from weeks to days.
- Traced 46% of team support tickets to an untyped delimited-text data pipeline and rebuilt it on Apache Parquet with Spark date and hour partitioning and streaming ingestion: backfills 10x faster, data freshness from 1 hour to 15 minutes, and reporting tickets down 77% year over year.
- Diagnosed a peak-season retry storm that cascaded into an authorization failure across 32 sites, authored the postmortem, and closed the detection gaps; those alarms have since caught multiple incidents before they reached customers. A CloudWatch threshold-tuning tool I built cut false-positive pages by over 90% and on-call interruptions from 5+ a week to 0 or 1.

### Machine Learning Engineer, RBC, Toronto — May 2019 – August 2019

- Built model-performance analytics in Python and SQL over Big Data on a Hadoop cluster, evaluated XGBoost and Random Forest with ROC curves and non-cumulative lifts, and implemented the LIME explainer for black-box model testing.

## PERSONAL PROJECTS

*Designed, built and operated solo, end to end: data model, API, client and the CDK infrastructure underneath.*

**Halvy** (React Native, TypeScript, AWS CDK, DynamoDB, Lambda). Shared-expense app for couples and households. Integer-cent split engine with exact settlement over an offline-first SQLite client and a serverless API. On the App Store and at usehalvy.com.

**MedSentri** (React, TypeScript, OpenSearch, API Gateway, Cognito, Claude). Normalizes FDA, Health Canada, ClinicalTrials.gov, WHO, CDC and RxNorm data into one searchable surface across roughly fifty Lambda handlers. Live at medsentri.com.

## EDUCATION

### University of Toronto Scarborough — September 2016 – April 2020

Honours Bachelor of Science, Mathematical and Computer Sciences

## SKILLS

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| **Languages** | TypeScript, Java, Python, JavaScript, SQL |
| **Distributed** | Circuit breakers, rate limiting, capacity modelling, chaos testing, multi-region design, alarm design, incident response |
| **AWS** | Lambda, DynamoDB, OpenSearch Serverless, ElastiCache/Valkey, API Gateway, Step Functions, Kinesis, CloudFront, CloudWatch, Bedrock |
| **Platform** | AWS CDK, CloudFormation, CI/CD design with automatic rollback, Apache Spark, Apache Parquet, streaming ingestion |
| **AI systems** | Bedrock (Claude), Model Context Protocol servers, agentic tool-use loops, sandboxed agent execution, multi-agent review pipelines |